

# Probability & Random Processes EE2011

Department of Electrical Engineering

**Correlation Coefficient,  
Mean and Variances of Linear Combination of Random Variables,  
Chebyshev Theorem**

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Although the covariance between two random variables does provide information regarding the nature of the relationship, the magnitude of  $\sigma_{XY}$  *does not indicate anything regarding the strength of the relationship*, since  $\sigma_{XY}$  is not scale-free.

Its magnitude will depend on the units used to measure both  $X$  and  $Y$ .

There is a scale-free version of the covariance called the **correlation coefficient** that is used widely in statistics.

#### Definition 4.5:

Let  $X$  and  $Y$  be random variables with covariance  $\sigma_{XY}$  and standard deviations  $\sigma_X$  and  $\sigma_Y$ , respectively. The correlation coefficient of  $X$  and  $Y$  is

$$\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}.$$

It should be clear to the reader that  $\rho_{XY}$  is free of the units of  $X$  and  $Y$ .

The correlation coefficient satisfies the inequality  $-1 \leq \rho_{XY} \leq 1$ .



It assumes a value of zero when  $\sigma_{XY} = 0$ .

Where there is an exact linear dependency, say  $Y \equiv a + bX$ ,  $\rho_{XY} = 1$  if  $b > 0$  and  $\rho_{XY} = -1$  if  $b < 0$ . (See Exercise 4.48.)

The correlation coefficient is the subject of more discussion in Chapter 12, where we deal with linear regression.



**Example 4.15:** Find the correlation coefficient between  $X$  and  $Y$  in Example 4.13.

**Solution:** Since

$$E(X^2) = (0^2) \left( \frac{5}{14} \right) + (1^2) \left( \frac{15}{28} \right) + (2^2) \left( \frac{3}{28} \right) = \frac{27}{28}$$

and

$$E(Y^2) = (0^2) \left( \frac{15}{28} \right) + (1^2) \left( \frac{3}{7} \right) + (2^2) \left( \frac{1}{28} \right) = \frac{4}{7},$$

we obtain

$$\sigma_X^2 = \frac{27}{28} - \left( \frac{3}{4} \right)^2 = \frac{45}{112} \text{ and } \sigma_Y^2 = \frac{4}{7} - \left( \frac{1}{2} \right)^2 = \frac{9}{28}.$$

Therefore, the correlation coefficient between  $X$  and  $Y$  is

$$\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y} = \frac{-9/56}{\sqrt{(45/112)(9/28)}} = -\frac{1}{\sqrt{5}}.$$



**Example 4.16:** Find the correlation coefficient of  $X$  and  $Y$  in Example 4.14.

**Solution:** Because

$$E(X^2) = \int_0^1 4x^5 dx = \frac{2}{3} \text{ and } E(Y^2) = \int_0^1 4y^3(1 - y^2) dy = 1 - \frac{2}{3} = \frac{1}{3},$$

we conclude that

$$\sigma_X^2 = \frac{2}{3} - \left(\frac{4}{5}\right)^2 = \frac{2}{75} \text{ and } \sigma_Y^2 = \frac{1}{3} - \left(\frac{8}{15}\right)^2 = \frac{11}{225}.$$

Hence,

$$\rho_{XY} = \frac{4/225}{\sqrt{(2/75)(11/225)}} = \frac{4}{\sqrt{66}}.$$

Note that although the covariance in Example 4.15 is larger in magnitude (disregarding the sign) than that in Example 4.16, the relationship of the magnitudes of the correlation coefficients in these two examples is just the reverse.

This is evidence that we cannot look at the magnitude of the covariance to decide on how strong the relationship is.



## 4.3 Means and Variances of Linear Combinations of Random Variables

We now develop some useful properties that will simplify the calculations of means and variances of random variables that appear in later chapters.

These properties will permit us to deal with expectations in terms of other parameters that are either known or easily computed.

All the results that we present here are valid for both discrete and continuous random variables.

Proofs are given only for the continuous case.

We begin with a theorem and two corollaries that should be, intuitively, reasonable to the reader.



### Theorem 4.5:

If  $a$  and  $b$  are constants, then

$$E(aX + b) = aE(X) + b.$$

#### *Proof:*

By the definition of expected value,

$$E(aX + b) = \int_{-\infty}^{\infty} (ax + b)f(x) dx = a \int_{-\infty}^{\infty} xf(x) dx + b \int_{-\infty}^{\infty} f(x) dx.$$

The first integral on the right is  $E(X)$  and the second integral equals 1.

Therefore, we have

$$E(aX + b) = aE(X) + b.$$



**Example 4.17:** Applying Theorem 4.5 to the discrete random variable  $f(X) = 2X - 1$ , rework Example 4.4 on page 115.

**Solution:** According to Theorem 4.5, we can write

$$E(2X - 1) = 2E(X) - 1.$$

Now

$$\begin{aligned}\mu = E(X) &= \sum_{x=4}^9 x f(x) \\ &= (4) \left( \frac{1}{12} \right) + (5) \left( \frac{1}{12} \right) + (6) \left( \frac{1}{4} \right) + (7) \left( \frac{1}{4} \right) + (8) \left( \frac{1}{6} \right) + (9) \left( \frac{1}{6} \right) = \frac{41}{6}.\end{aligned}$$

Therefore,

$$\mu_{2X-1} = (2) \left( \frac{41}{6} \right) - 1 = \$12.67,$$

as before.



**Example 4.18:** Applying Theorem 4.5 to the continuous random variable  $g(X) = 4X + 3$ , rework Example 4.5 on page 115.

**Solution:** For Example 4.5, we may use Theorem 4.5 to write

$$E(4X + 3) = 4E(X) + 3.$$

Now

$$E(X) = \int_{-1}^2 x \left( \frac{x^2}{3} \right) dx = \int_{-1}^2 \frac{x^3}{3} dx = \frac{5}{4}.$$

Therefore,

$$E(4X + 3) = (4) \left( \frac{5}{4} \right) + 3 = 8,$$

as before.



### Theorem 4.6:

The expected value of the sum or difference of two or more functions of a random variable  $X$  is the sum or difference of the expected values of the functions. That is,

$$E[g(X) \pm h(X)] = E[g(X)] \pm E[h(X)].$$

### *Proof:*

By definition,

$$\begin{aligned} E[g(X) \pm h(X)] &= \int_{-\infty}^{\infty} [g(x) \pm h(x)]f(x) \, dx \\ &= \int_{-\infty}^{\infty} g(x)f(x) \, dx \pm \int_{-\infty}^{\infty} h(x)f(x) \, dx \\ &= E[g(X)] \pm E[h(X)]. \end{aligned}$$



**Example 4.19:** Let  $X$  be a random variable with probability distribution as follows:

|        |               |               |   |               |
|--------|---------------|---------------|---|---------------|
| $x$    | 0             | 1             | 2 | 3             |
| $f(x)$ | $\frac{1}{3}$ | $\frac{1}{2}$ | 0 | $\frac{1}{6}$ |

Find the expected value of  $Y = (X - 1)^2$ .

**Solution:** Applying Theorem 4.6 to the function  $Y = (X - 1)^2$ , we can write

$$E[(X - 1)^2] = E(X^2 - 2X + 1) = E(X^2) - 2E(X) + E(1).$$

From Corollary 4.1,  $E(1) = 1$ , and by direct computation,

$$E(X) = (0)\left(\frac{1}{3}\right) + (1)\left(\frac{1}{2}\right) + (2)(0) + (3)\left(\frac{1}{6}\right) = 1 \text{ and}$$

$$E(X^2) = (0)\left(\frac{1}{3}\right) + (1)\left(\frac{1}{2}\right) + (4)(0) + (9)\left(\frac{1}{6}\right) = 2.$$

Hence,

$$E[(X - 1)^2] = 2 - (2)(1) + 1 = 1.$$



**Example 4.20:** The weekly demand for a certain drink, in thousands of liters, at a chain of convenience stores is a continuous random variable  $g(X) = X^2 + X - 2$ , where  $X$  has the density function

$$f(x) = \begin{cases} 2(x-1), & 1 < x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the expected value of the weekly demand for the drink.

**Solution:** By Theorem 4.6, we write

$$E(X^2 + X - 2) = E(X^2) + E(X) - E(2).$$

From Corollary 4.1,  $E(2) = 2$ , and by direct integration,

$$E(X) = \int_1^2 2x(x-1) dx = \frac{5}{3} \text{ and } E(X^2) = \int_1^2 2x^2(x-1) dx = \frac{17}{6}.$$

Now

$$E(X^2 + X - 2) = \frac{17}{6} + \frac{5}{3} - 2 = \frac{5}{2},$$

so the average weekly demand for the drink from this chain of efficiency stores is 2500 liters.



### Theorem 4.7:

The expected value of the sum or difference of two or more functions of the random variables  $X$  and  $Y$  is the sum or difference of the expected values of the functions. That is,

$$E[g(X, Y) \pm h(X, Y)] = E[g(X, Y)] \pm E[h(X, Y)].$$

**Proof:** By Definition 4.2,

$$\begin{aligned} E[g(X, Y) \pm h(X, Y)] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} [g(x, y) \pm h(x, y)] f(x, y) \, dx \, dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f(x, y) \, dx \, dy \pm \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x, y) f(x, y) \, dx \, dy \\ &= E[g(X, Y)] \pm E[h(X, Y)]. \end{aligned}$$

### Corollary 4.3:

Setting  $g(X, Y) = g(X)$  and  $h(X, Y) = h(Y)$ , we see that

$$E[g(X) \pm h(Y)] = E[g(X)] \pm E[h(Y)].$$



### Corollary 4.4:

Setting  $g(X, Y) = X$  and  $h(X, Y) = Y$ , we see that

$$E[X \pm Y] = E[X] \pm E[Y].$$

If  $X$  represents the daily production of some item from machine  $A$  and  $Y$  the daily production of the same kind of item from machine  $B$ , then  $X + Y$  represents the total number of items produced daily by both machines.

Corollary 4.4 states that the average daily production for both machines is equal to the sum of the average daily production of each machine.



### Theorem 4.8:

Let  $X$  and  $Y$  be two independent random variables. Then

$$E(XY) = E(X)E(Y).$$

**Proof:** By Definition 4.2,

$$E(XY) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f(x, y) \, dx \, dy.$$

Since  $X$  and  $Y$  are independent, we may write

$$f(x, y) = g(x)h(y),$$

where  $g(x)$  and  $h(y)$  are the marginal distributions of  $X$  and  $Y$ , respectively.

Hence,

$$\begin{aligned} E(XY) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy g(x)h(y) \, dx \, dy = \int_{-\infty}^{\infty} xg(x) \, dx \int_{-\infty}^{\infty} yh(y) \, dy \\ &= E(X)E(Y). \end{aligned}$$



### Corollary 4.5:

Let  $X$  and  $Y$  be two independent random variables. Then  $\sigma_{XY} = 0$ .

#### *Proof:*

The proof can be carried out by using Theorems 4.4 and 4.8. └

Theorem 4.8 can be illustrated for discrete variables by considering the experiment of tossing a green die and a red die.

Let the random variable  $X$  represent the outcome on the green die and the random variable  $Y$  represent the outcome on the red die.

Then  $XY$  represents the product of the numbers that occur on the pair of dice.

In the long run, the average of the products of the numbers is equal to the product of the average number that occurs on the green die and the average number that occurs on the red die.



**Example 4.21:** It is known that the ratio of gallium to arsenide does not affect the functioning of gallium-arsenide wafers, which are the main components of microchips. Let  $X$  denote the ratio of gallium to arsenide and  $Y$  denote the functional wafers retrieved during a 1-hour period.  $X$  and  $Y$  are independent random variables with the joint density function

$$f(x, y) = \begin{cases} \frac{x(1+3y^2)}{4}, & 0 < x < 2, 0 < y < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Show that  $E(XY) = E(X)E(Y)$ , as Theorem 4.8 suggests.

**Solution:** By definition,

$$E(XY) = \int_0^1 \int_0^2 \frac{x^2 y (1 + 3y^2)}{4} dx dy = \frac{5}{6}, \quad E(X) = \frac{4}{3}, \quad \text{and} \quad E(Y) = \frac{5}{8}.$$

Hence,

$$E(X)E(Y) = \left(\frac{4}{3}\right) \left(\frac{5}{8}\right) = \frac{5}{6} = E(XY).$$



## 4.4 Chebyshev's Theorem

In Section 4.2 we stated that the variance of a random variable tells us something about the variability of the observations about the mean. If a random variable has a small variance or standard deviation, we would expect most of the values to be grouped around the mean. Therefore, the probability that the random variable assumes a value within a certain interval about the mean is greater than for a similar random variable with a larger standard deviation. If we think of probability in terms of area, we would expect a continuous distribution with a large value of  $\sigma$  to indicate a greater variability, and therefore we should expect the area to be more spread out, as in Figure 4.2(a). A distribution with a small standard deviation should have most of its area close to  $\mu$ , as in Figure 4.2(b).



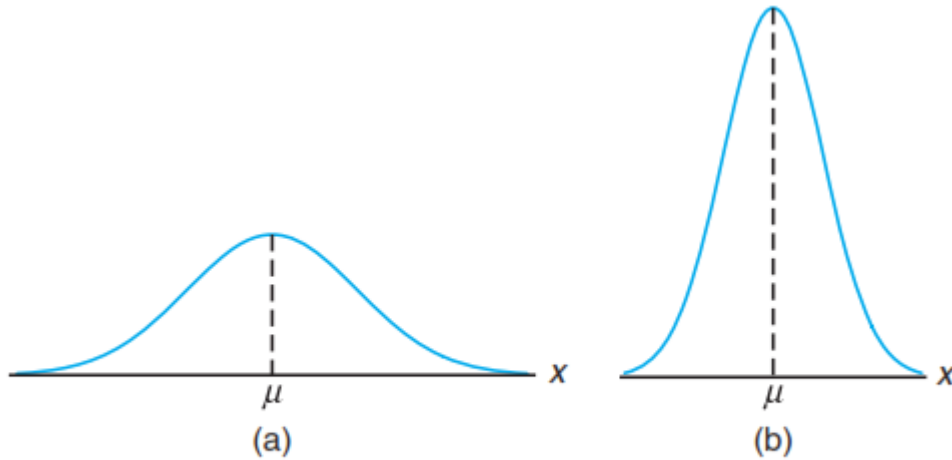


Figure 4.2: Variability of continuous observations about the mean.

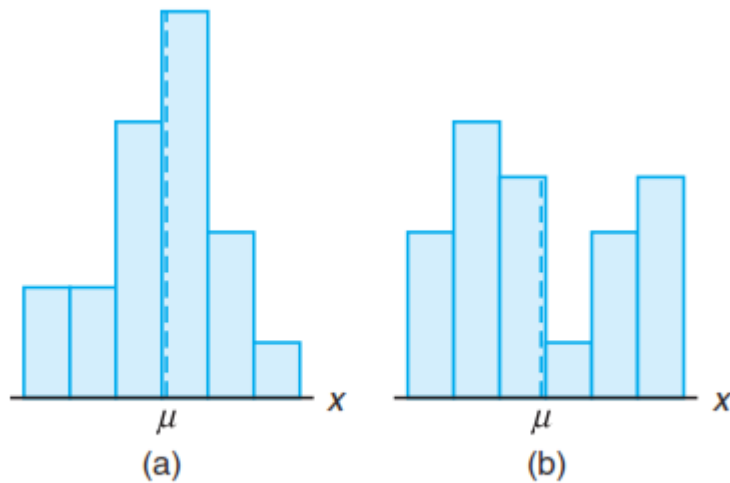


Figure 4.3: Variability of discrete observations about the mean.

We can argue the same way for a discrete distribution. The area in the probability histogram in Figure 4.3(b) is spread out much more than that in Figure 4.3(a) indicating a more variable distribution of measurements or outcomes.

The Russian mathematician P. L. Chebyshev (1821–1894) discovered that the fraction of the area between any two values symmetric about the mean is related to the standard deviation. Since the area under a probability distribution curve or in a probability histogram adds to 1, the area between any two numbers is the probability of the random variable assuming a value between these numbers.

The following theorem, due to Chebyshev, gives a conservative estimate of the probability that a random variable assumes a value within  $k$  standard deviations of its mean for any real number  $k$ .



**Theorem 4.10:** **(Chebyshev's Theorem)** The probability that any random variable  $X$  will assume a value within  $k$  standard deviations of the mean is at least  $1 - 1/k^2$ . That is,

$$P(\mu - k\sigma < X < \mu + k\sigma) \geq 1 - \frac{1}{k^2}.$$

For  $k = 2$ , the theorem states that the random variable  $X$  has a probability of at least  $1 - 1/2^2 = 3/4$  of falling within two standard deviations of the mean. That is, three-fourths or more of the observations of any distribution lie in the interval  $\mu \pm 2\sigma$ . Similarly, the theorem says that at least eight-ninths of the observations of any distribution fall in the interval  $\mu \pm 3\sigma$ .



**Example 4.27:** A random variable  $X$  has a mean  $\mu = 8$ , a variance  $\sigma^2 = 9$ , and an unknown probability distribution. Find

(a)  $P(-4 < X < 20)$ ,

(b)  $P(|X - 8| \geq 6)$ .

**Solution:** (a)  $P(-4 < X < 20) = P[8 - (4)(3) < X < 8 + (4)(3)] \geq \frac{15}{16}$ .

(b)  $P(|X - 8| \geq 6) = 1 - P(|X - 8| < 6) = 1 - P(-6 < X - 8 < 6)$   
 $= 1 - P[8 - (2)(3) < X < 8 + (2)(3)] \leq \frac{1}{4}$ .



Chebyshev's theorem holds for any distribution of observations, and for this reason the results are usually weak. The value given by the theorem is a lower bound only. That is, we know that the probability of a random variable falling within two standard deviations of the mean can be *no less* than  $3/4$ , but we never know how much more it might actually be.

Only when the probability distribution is known can we determine exact probabilities. For this reason we call the theorem a *distribution-free* result. When specific distributions are assumed, as in future chapters, the results will be less conservative. The use of Chebyshev's theorem is relegated to situations where the form of the distribution is unknown.

